

```

linux:/opt/nayana/prgms/core_dir # ulimit -c
unlimited
linux:/opt/nayana/prgms/core_dir # echo core %p %u %g %s %t %h %e %c > /proc/sys/kernel/core_pattern
linux:/opt/nayana/prgms/core_dir # cat /proc/sys/kernel/core_pattern
core %p %u %g %s %t %h %e %c
linux:/opt/nayana/prgms/core_dir # ../test
Segmentation fault (core dumped)
linux:/opt/nayana/prgms/core_dir # ls
core_4861_0_0_11_1347438689_linux_test_18446744073709551615
linux:/opt/nayana/prgms/core_dir #

```

Figure 1: Set the template for the core file name

the message ‘Segmentation fault (core dumped)’. By default, the core dump file name will be core. You can also format the dump file name by defining a template in `/proc/sys/kernel/core_pattern`, using % specifiers that are substituted as follows:

%	a single % character in the core file name
%p	PID of the dumped process
%u	(numeric) real UID of the dumped process
%g	(numeric) real GID of the dumped process
%s	number of the signal causing the dump
%t	The time of the dump, expressed as seconds since the Epoch (00:00h, 1 Jan 1970, UTC)
%h	Hostname
%e	executable filename
%c	core file size soft resource limit of crashing process

Figure 1 shows how I set the template for the file name, ran our test program, and got a core file with a name as per the specified pattern.

If `/proc/sys/kernel/core_pattern` does not include %p but `/proc/sys/kernel/core_uses_pid` is set to 1, then the core file name will be appended with .PID (the PID of the crashing process). If %p is in `core_pattern` then the `core_uses_pid` value is ignored.

You can also dump the core of a running process (that has not crashed) using the `gcore` utility, specifying the PID of the process. For example, if a program is started under the PID 5004, then `gcore 5004` will dump a core file for the running process, without a crash—it continues running.

Debugging using core dumps

Core dumps can be debugged using GDB. When gdb is run with the binary that created the core file, as well as the core file, it reads the contents of the core dump, and understands the following:

- The name of the process that dumped the core file
- The signal sent to the process to dump the core file
- Any illegal statement in the program, with the line number
- Values of variables
- Values of registers, etc.

Figure 2 shows a sample run of GDB in this manner. This can be viewed at http://linuxforu.com/article_source_code/dec12/coredump.zip.

Reducing the core file size

There might be space restrictions per user on a multi-user system; large core files, especially when multi-threaded applications crash (since they would use more memory), might just eat up available storage. Linux supports the reduction of the core file size using the filter option (`/proc/<pid>/coredump_filter`), which can be used to dump only specific memory to the core file when the process crashes. Table 1 shows what each bit signifies—what memory to dump:

Table 1

Bit	What to dump
0	Anonymous private mappings
1	Anonymous shared mappings
2	File-backed private mappings
3	File-backed shared mappings
4	ELF headers
5	Huge TLB private
6	Huge TLB shared

The default value of `coredump_filter` is 0x3, which dumps only anonymous, private and shared mappings of the crashed memory. The file accepts hex, decimal and octal values.

The sample use of the `coredump_filter` can be viewed at http://linuxforu.com/article_source_code/dec12/coredump.zip.

You can obtain the memory records of a crashed process, which can help in debugging the program. You have the flexibility to specify the core file name pattern and to optimise size using `coredump_pattern` and `coredump_filter`. 

Useful links

- [1] http://en.wikipedia.org/wiki/Core_dump
- [2] <http://linux.die.net/man/5/core>
- [3] http://www.akadia.com/services/ora_enable_core.html
- [4] <http://www.cyberciti.biz/tips/linux-core-dumps.html>
- [5] <http://stackoverflow.com/questions/17965/generate-a-core-dump-in-linux>

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